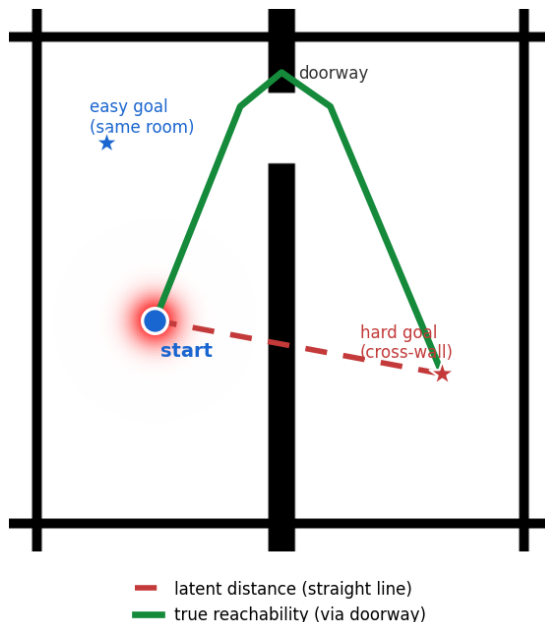


Progress report — Value Learning

A model-based agent plans by *imagining* the future: it rolls a learned world-model forward over many candidate action sequences and keeps the one whose imagined endpoint $\hat{z}_{t+H}$ — a vector, the model’s compressed “state,” or *latent* — is **closest** to the goal z_g . Everything hinges on what “closest” means: the **plan score** we assign to each candidate’s imagined endpoint $(\hat{z}_{t+H}, z_g)$.



The benchmark, TwoRoom: two rooms joined by one doorway. *Easy* goals sit in the same room; *hard* goals are across the wall. The **straight latent distance** points through the wall, while **true reachability** must detour through the doorway.

The default plan score — and why it is wrong

Today’s latent world-model planners — LeWM, DINO-WM, PLDM — all score closeness with the *latent distance* $\|\hat{z}_{t+H} - z_g\|^2$ (plain Euclidean distance in latent space). **The catch: this distance is never trained to represent actual distance.** It simply falls out of a model trained to *predict frames*, so nothing forces it to track *how far to walk*. A goal on the far side of a wall looks “near” in latent space yet is many steps away — the straight line cuts through the wall (above), so the planner steers into it. We test four plan scores on this benchmark, on easy and hard goals, giving every one the *same* planning budget; we report the % of 100 goals reached.

The four plan scores we compare

- **Latent distance (Default)** = $\|\hat{z} - z_g\|$ — the current standard. Free, no training; completely blind to obstacles.
- **Regression** — “TRM” (others) — the prior published metric: a network trained (supervised) to predict the step-gap between two states drawn from logged trajectories. Better than raw distance, but it only knows the paths the data *happened* to take — if they wandered, it learns inflated distances, and it cannot infer a shortcut it never saw.
- **TD value (ours)** — *value learning*. Instead of copying observed gaps, learn a goal-conditioned *value* $d(z, z_g) \approx \text{steps-to-go}$ with a Bellman / temporal-difference update, $d(z, g) \leftarrow 1 + \min_{z'} d(z', g)$.

The min/bootstrap **stitches** short transitions into the globally *shortest* path, so even from non-optimal data it recovers true reachability and finds the route through the doorway. (Our one-line fix lives here: the bootstrap must be *optimistic* — take the shortest stitched path, not the average.)

- **Online TD (ours) — *self-practice*.** Let the planner act with the current value d , collect those rollouts, and keep training d on them — so the score becomes accurate exactly on the states the planner actually visits (which a fixed log under-covers).

Results: % of goals reached

		Latent dist.	TRM	TD value	Online TD
World model	Goals	(Default)	(others)	(ours)	(ours)
DINO-WM	easy	71	58	95	95
DINO-WM	hard	34	29	83	93
LeWM	easy	78	62	92	90
LeWM	hard	16	29	64	68

Green = best plan score in that row; *red* = the default score collapsing on hard goals. A perfect-knowledge planner scores $\approx 100\%$.

What it means

- **On hard goals the default score all but fails (16–34%),** and the prior metric (TRM) barely helps (29%, even trailing the default on DINO) — copying observed gaps isn’t the same as finding shortest paths.
- **Value learning (TD) is the jump** (hard: 16 $\rightarrow$ 64, 34 $\rightarrow$ 83). And the *optimistic-target fix* alone roughly doubled hard-goal success (52 $\rightarrow$ 64, 65 $\rightarrow$ 83) — a one-line change to the training objective.
- **Self-practice adds the last mile** (DINO 83 $\rightarrow$ **93**, LeWM 64 $\rightarrow$ **68**) — and it is the practice *data*, not extra compute: training the offline value 2 $\times$ longer instead over-fits and gets *worse* (83 $\rightarrow$ 73, 64 $\rightarrow$ 60).

Details. Hard / easy = the n100 goal set at geodesic 90–180 / 20–70 px (50 cross-wall + 50 same-room). Matched budget = CEM $30 \times 5 \approx 150$ candidate evaluations. TD value = offline goal-conditioned temporal-difference on a quasimetric head; the “optimistic” setting is the TD *expectile* ($\tau < 0.5$, biased toward the shortest path). Self-practice = the same value continued online on planner-collected rollouts; the gain was verified against a matched-compute control.